Delta Hedging a Binary Option with a Perpetual

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1 The Core Idea

A "Bitcoin above K" binary pays \$1 if BTC settles above K and nothing otherwise. Holding 1,000 of them is a bet on one number. A risk manager wants to know how much of that bet can be sold back to the market, and what it costs to do so. The repository buys the contracts on Polymarket and hedges the BTC exposure with the Hyperliquid perpetual.

- **Delta explodes near expiry:** for a binary at the strike, delta grows like $1/\sqrt{T}$ and gamma is of order $1/T$ within a few $\sigma\sqrt{T}$ of the strike, so the hedge needs ever larger corrections close to settlement.
- **Costs decide the result:** each rebalance pays the taker fee and crosses the spread, so a hedge that tracks delta closely can lose more than one that tracks it loosely.

The same question is then asked of a strip of Kalshi binaries and of Deribit options over 207 daily expiries (2026-03-19 to 2026-10-11).

2 Why It Matters

- **A linear hedge cannot cover a jump payoff.** The perpetual removes the first-order exposure to S only. At settlement the binary pays 0 or \$1,000 whatever the hedge did, and the loss from falling implied volatility and time decay stays unhedged. In the live session the Polymarket leg lost \$131.22 while the hedge earned \$77.44.
- **Delta is unstable near the strike.** For 1,000 shares at $S = K$, $\sigma = 0.40$ (illustrative), the hedge is 0.454 BTC with 6 hours left, 1.111 BTC with 1 hour left and 2.223 BTC with 15 minutes left.
- **Tracking costs money.** The dynamic hedge in the live session traded \$403,188 of perp notional against \$38,003 for the static one, and paid \$181.43 in fees against \$5.70.
- **A hedge lowers risk, not the carry.** Over 207 expiries the Kalshi strip halves the daily P&L standard deviation, yet every structure loses on average. The hedge removes variance, not cost.

3 Assumptions

Every assumption below is built into the code. For each one, the last sentence says what moves if it is wrong.

- **Pricing model.** Black–Scholes digital with zero rate, lognormal S, constant σ over the life of the contract, T in 365-day years. Jumps fatten the tails, so the true probability of ending far from K is higher than $N(d_2)$, and the delta near the strike is overstated.
- **Perp equals the settlement index.** Polymarket resolves on a Binance BTC/USDT print and Kalshi on the BRTI index, while the hedge is a Hyperliquid perp. The code treats them as the same asset. Any basis between them is unhedged and invisible in the results.
- **Volatility source.** The live engine takes σ from the Polymarket mid (held 300 s, then DVOL). The backtests take DVOL at entry or the median Kalshi implied vol. The live σ comes from the market being hedged, so hedge and position share the same error. A DVOL that is too high lowers delta and the hedge ratio.
- **Execution.** Taker orders sweep the published book level by level with no persistent impact, and orders fill on the same snapshot that triggered them (no latency). Maker orders fill only when a trade crosses their price strictly, with no queue position. Polymarket P&L is marked at the mid, not at the bid. Real fills would be worse for large or fast orders and slower for passive ones.
- **Fees and funding.** Hyperliquid base tier (taker 4.5 bp, maker 1.5 bp), Polymarket crypto taker rate 0.07 and maker rate 0 with no rebate, Kalshi taker rate 0.07 rounded up to the cent, Deribit $\min(0.0003\ \text{BTC}, 12.5\%)$ per leg with the combo rule not applied. A volume tier or a rebate would lower the cost of the dynamic hedge most.
- **Backtest window.** Each day runs from 21:00 UTC on day $D-1$ to 21:00 UTC on day D. The perp hedge is sized once at entry and closed at the end, with no rebalancing in between. The backtest measures a static hedge, not a dynamic one.
- **Strip construction.** Equal numbers of NO contracts on strikes aimed at \$250 intervals, with the premium of the single strike: nine legs on 177 of the 207 days, five on the other 30 where Kalshi lists fewer strikes. One common σ feeds the aggregate delta. A different weighting or a wider strip would change the shape of the payoff.
- **Independent days.** The t-tests and the bootstrap treat the 207 days as independent draws. Consecutive days share a volatility regime, so the standard errors are probably too small.
- **Sample.** One live session of 87 minutes with one strike (\$84,000) and 1,000 shares, then seven months of daily expiries in one market regime. Seven days move more than 5%.

4 Method

4.1 Pricing and Greeks

With zero rate, the code (`core/math_engine.py`) prices the binary as a cash-or-nothing call:

$$d_2 = \frac{\ln(S/K) - \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}, \qquad V = N(d_2), \tag{1}$$

$$\Delta = \frac{\varphi(d_2)}{S\sigma\sqrt{T}}, \qquad \Gamma = -\frac{\varphi(d_2)\, d_1}{S^2\sigma^2 T}, \quad d_1 = d_2 + \sigma\sqrt{T}. \tag{2}$$

Delta is a bell curve in S centred just above K, with a width of order $\sigma\sqrt{T}$ and a height of order $1/(S\sigma\sqrt{T})$. A shorter life or a lower σ makes the bell taller and narrower: a larger hedge, in a

thinner band around the strike. The implied volatility is the root of $N(d_2(\sigma)) = V$ on $[10^{-3}, 5]$. For $S < K$ the price peaks at $N(-\sqrt{2|\ln(S/K)|}) < 0.5$ (0.461 in the worked example below), so the code keeps the low-volatility root and returns nothing above the peak.

4.2 Volatility Input

The engine takes σ from one of three sources: the volatility implied by the Polymarket mid, the Deribit DVOL index, or a time-weighted EWMA of Hyperliquid returns. The EWMA decays its sums by $w = e^{-\ln 2\,\Delta t/1800}$ at each 60-second step and returns

$$\hat{\sigma} = \sqrt{\frac{\sum w\, r^2}{\sum w\, \Delta t} \cdot 365 \cdot 86{,}400}. \tag{3}$$

The half-life is 1,800 s. The last valid implied value is held for 300 s, then the engine falls back to DVOL, then to the EWMA. In the live session the implied value was available at all 4,880 marks. It started at 0.332 against a DVOL of 0.379 and ended at 0.305.

4.3 Hedge Rules

For Q YES shares held, the target perp position in BTC is

$$h = -Q\,\Delta(S, K, \sigma, T). \tag{4}$$

Two strategies are compared:

- **Static:** one limit order on the passive side of the Hyperliquid book, sized to the initial h_0. It is a maker order and fills only when a trade crosses its price. In the session it sold 0.4542 BTC at an average of \$83,671.0 in 40 partial fills. The fills run from 23.1 to 61.6 seconds after entry (38.5 seconds).
- **Dynamic:** a market order whenever $|h - \text{position}|$ exceeds $\max\big(\text{lot}, \text{min notional}/S\big)$. With a \$10 minimum notional the threshold is about 1.2×10^{-4} BTC, so small delta moves trigger orders. The threshold is the only tuning parameter, and it sits at the exchange minimum.

A kill switch closes all positions at expiry minus 2 hours (14:00 UTC for the 16:00 UTC expiry), before the gamma region. The trade-off is explicit: it gives up the last two hours of protection to avoid the region where hedging is both most needed and most expensive.

4.4 Execution and Fees

Taker orders walk the order book level by level, and each level is a separate fill. There is no persistent market impact. Fees are those in `risk/fees.py`:

$$\text{fee}_{HL} = |N| \cdot b/10^4, \quad b = 4.5 \text{ (taker)},\ 1.5 \text{ (maker)}, \qquad \text{fee}_{PM} = C \cdot 0.07 \cdot p(1-p). \tag{5}$$

Here N is the trade notional, C the number of shares and p the share price. The Polymarket fee is largest at $p = 0.5$ and falls to zero at 0 and 1. The ledger charges funding at each UTC hour boundary as position $\times\, S \times$ the last hourly rate.

4.5 Kalshi Strip

A single Kalshi NO binary has the same exploding gamma. The strip buys NO contracts on up to 9 strikes K_i (9 on 177 days, 5 on 30), with n_i contracts at price p_i each, and the same premium budget B as the single strike (1,000 contracts at the at-the-money strike):

$$n_i = \text{round}\Big(\frac{B}{\sum_j p_j}\Big), \qquad h = -\sum_i n_i\,\Delta_{\text{NO}}(K_i), \quad \Delta_{\text{NO}}(K) = -\frac{\varphi(d_2(K))}{S\sigma\sqrt{T}}. \tag{6}$$

The payoff $\sum_i n_i \mathbf{1}\{S_T < K_i\}$ is a staircase. With $n_i = Q\,\Delta K$ it matches the put spread $Q[(K_{\max} - S_T)^+ - (K_{\min} - S_T)^+]$ within $Q\,\Delta K$ in every scenario. The gamma of the strip is that of a vanilla put spread: it diverges like $1/\sqrt{T}$ instead of $1/T$ and is spread over nine strikes. The strip hedge is one static perp order, closed at the end of the window.

4.6 Deribit Benchmark

The benchmark buys a put spread and a vanilla put with the same payoff per dollar of BTC move ($q = n$/step BTC) and the same static perp hedge. Prices come from Deribit trades in the three hours before entry (a linear smile in log-moneyness, plus or minus the measured half-spread), because no historical book exists. The fee is $q \cdot \min(0.0003\text{ BTC}, 12.5\%\text{ of premium})$ per leg, at entry and at exit.

5 Method Comparison

Method	Strength	Weakness
Static perp hedge	One maker order, 1.5 bp, few fees	Delta drifts away from the initial hedge
Dynamic perp hedge	Net delta stays near zero	Taker fee on every rebalance, 4.5 bp
Kalshi strip	Gamma spread over nine strikes	Wide Kalshi spreads, no strikes below the grid
Deribit put spread	Bounded gamma, tighter spreads	Strike grid and 0.1 BTC increments; priced from trades

The static hedge suits a short holding period with small moves, where the residual delta is small. The dynamic hedge pays for its fees only where gamma is large and an open delta would lose more than the fee. The strip and the options change the position rather than the hedge, and the choice between them comes down to which venue has the tighter spread.

6 Results

6.1 Live Session

The recorded session of 12 October 2026 lasted 87 minutes, from 08:37 to 10:04 UTC ($K =$ \$84,000, $Q = 1000$ shares). Amounts are in dollars and are the final marks of the journal. Every column is a contribution to the total: costs are negative, funding received is positive. The next two tables use the same convention.

Strategy	PM P&L	HL P&L	PM fees	HL fees	Funding	Total
Static	−131.22	+77.44	−16.05	−5.70	+0.73	−74.80
Dynamic	−131.22	+73.19	−16.05	−181.43	+0.72	−254.80

Figures are rounded to the cent, so a row can miss its total by one cent: the dynamic components add up to −254.79 (unrounded total −254.80).

The perp P&L is similar in both cases, so the gap of about $176 in Hyperliquid fees decides the ranking. It comes from taker fees on the dynamic hedge: 1,548 rebalancing fills against 40 static fills. The dynamic hedge does what it is built to do. Its mean absolute net delta is 0.0001 BTC against 0.0333 BTC for the static hedge, about 300 times tighter. That precision bought nothing here: S stayed between $83,271.5 and $83,696.5, and the static hedge earned $4.25 more on the perp leg.

The Polymarket leg explains most of the loss. The 1,000 shares were bought at an average of 0.3562 against a mid of 0.34, a cost of $16.22 on top of the $16.05 fee, and the mid then fell to 0.225. The hedge covers the part of that fall caused by the $170 drop in BTC (about $0.454 \times 170 = \$77$). The rest, about $38, is time decay and the fall in implied volatility from 0.332 to 0.305. No perp hedge can recover that.

The session does not show the last hours before expiry, when gamma is large: it ended with almost six hours left.

6.2 One Kalshi Trade

The first backtest buys 1,000 NO contracts at the $82,999.99 strike (price 0.51, premium $510) for the 2026-10-11 expiry, against a strip with the same budget. Kalshi P&L is net of the Kalshi fee. BTC settled at $83,543.76.

Structure	Kalshi P&L	Perp P&L	HL fees	Funding	Total
Single strike	−527.50	+162.36	−18.70	+1.52	−382.32
Strip (9 legs)	−297.70	+152.59	−17.58	+1.43	−161.25

The strip does better by $221, almost all of it on the Kalshi leg. Settlement above the single strike made every NO worthless, while the two strip strikes above $83,543.76 (113 contracts each) paid $226. Both hedges are long perp, so the rise in BTC helped both. One day, one direction: the result says a strip pays when BTC ends near its strikes, not that it is better on average.

6.3 207 Daily Expiries

Hedged P&L per daily expiry, net of Kalshi fees, Hyperliquid fees and funding. σ is the DVOL value.

Structure	Mean ($)	Std ($)	Winning days (%)
Kalshi single strike	−85.01	361.80	41.1
Kalshi strip	−57.98	185.95	31.9
Deribit put spread	−36.3	133.0	25.1
Deribit vanilla put	−42.9	175.9	27.1

- **Risk:** the strip cuts the standard deviation of daily P&L by about half (ratio 0.514).
- **Mean:** the strip improves the mean by $27.03 per day, but the paired test gives $p = 0.136$. This is not a significant gain.

- **Deribit:** the put spread beats the strip by \$21.7 per day ($p = 0.002$). The sign reverses in September–October (−34.5 per day) when the Kalshi spread narrows.

The strip lowers the standard deviation because it pays something on most days (82% of days against 47% for the single strike), where the single strike pays 1,000 or nothing. It wins on fewer days (31.9% against 41.1%) because its payoff is smaller when it does pay.

The mean decomposes as follows (dollars per day, same backtest):

Structure	Kalshi	Perp	HL fees	Funding	Total
Single strike	−84.23	+19.57	−17.25	−3.11	−85.01
Strip	−53.78	+14.12	−15.52	−2.80	−57.98

The Kalshi leg carries the loss (funding is a net cost here because the long perp pays): the mean fee is \$16.69 for the single strike and \$13.67 for the strip, and the entry spread and the gap between the quoted price and the realised frequency make up the rest. The perp gains because the hedge is long on all 207 days and BTC rose on average over these windows (mean daily log return +0.078%, standard deviation 1.92%). That gain is drift, not skill, and it would turn negative in a falling market. Annualised, the realised volatility of these windows is 36.6% against a mean DVOL of 40.3%, so the volatility used for delta was a little above what BTC delivered.

Period	Days	Strip − single	p	Put spread − strip	p
March–May	74	−2.29	0.936	+49.6	< 0.001
June–August	92	+32.87	0.216	+24.2	0.009
September–October	41	+66.82	0.164	−34.5	0.008

The strip gains more as the sample moves forward, but no period is significant on its own. The Deribit gap follows the Kalshi spread, which fell from \$71.4 per day in March–May to \$11.1 in September–October: when Kalshi was expensive, Deribit won, and when it became cheap, the strip won. The reversal survives dropping the extreme days (−32.0 per day, $p = 0.012$, 38 days). Deribit is priced from a fitted smile, so this is a model against a quoted market as much as two venues.



Figure 1: Live session: BTC price, binary delta against the two hedges, and net delta.

7 Worked Example

The inputs are illustrative: $S = 83{,}600$, $K = 84{,}000$, $\sigma = 0.40$, $T = 6$ hours, $Q = 1000$ shares.

Step 1. $T = 6 \cdot 3600/(365 \cdot 24 \cdot 3600) = 6.849 \times 10^{-4}$ years, so $\sigma\sqrt{T} = 0.010468$.

Step 2. $d_2 = \big(\ln(83600/84000) - \frac{1}{2} \cdot 0.010468^2\big)/0.010468 = -0.4612$, and $V = N(-0.4612) = 0.3223$.

Step 3. $\Delta = \varphi(-0.4612)/(83600 \cdot 0.010468) = 4.0986 \times 10^{-4}$ per share. The target hedge is $h = -1000 \cdot \Delta = -0.4099$ BTC, a short of about \$34,264.

Step 4. At the taker fee the entry costs $34{,}264 \cdot 4.5/10^4 = \$15.42$. As a maker order the fee is $34{,}264 \cdot 1.5/10^4 = \$5.14$.

Step 5. If S moves to 83,700, the delta becomes 0.4287 BTC. The dynamic hedge sells another 0.01884 BTC (\$1,577), which costs \$0.71 in taker fees: one rebalance is cheap, and the cost comes from their number.

Sensitivity. Doubling σ to 0.80 cuts the hedge from 0.4099 to 0.2215 BTC and raises the binary price from 0.3223 to 0.4058. Halving it to 0.20 gives 0.6001 BTC. Only $\sigma\sqrt{T}$ matters, so $\sigma = 0.80$ with 6 hours gives the same numbers as $\sigma = 0.40$ with 24 hours. Time matters more than volatility, and it acts differently on each side of the strike (hedge in BTC for 1,000 shares):

Time left	24 h	6 h	2 h	1 h	15 min
$S = 83{,}600$	0.2215	0.4099	0.5766	0.5970	0.1838
$S = 84{,}000$	0.2268	0.4537	0.7858	1.1113	2.2226

Below the strike the hedge rises, then collapses as the contract becomes worthless (price 0.0127 at 15 minutes). At the strike it keeps rising. With 15 minutes left, a \$100 move at the strike changes the hedge by 0.320 BTC (about \$26,900) and one rebalance costs \$12.12 in taker fees, against \$0.12 with 6 hours left.

8 Limits and Extensions

- **One live session:** the static against dynamic comparison rests on 87 minutes in a quiet range, with no gamma region. The kill-switch checks that need the session to reach expiry are pending.
- **Model risk:** the digital price depends on σ, and for $S < K$ the implied volatility has no root when the mid is above the price peak.

Three extensions would address these limits. First, a band hedge that trades only when the net delta leaves a band widening with the fee, which keeps most of the tracking and cuts the number of fills. Second, a dynamic-hedge backtest on the 207 days using recorded Hyperliquid books, so the comparison no longer rests on one session. Third, a smile-aware delta, since implied volatility fell from 0.332 to 0.305 in 87 minutes.

9 Conclusion

A perpetual hedge turns a binary into a position whose remaining risk is volatility, time decay and fees, and the live session shows that fees can be the largest of the three. Hedge as loosely as the open delta allows, and near expiry change the position rather than the hedge.